

EXPERIMENT NO. 1

Aim:

To analyze, specify and document the functional and non-functional requirements of an Online Vehicle Rental System using the IEEE Software Requirements Specification (SRS) standard.

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ONLINE VEHICLE RENTAL SYSTEM

Problem Statement

Traditional vehicle rental models depend heavily on localized brick-and-mortar operations, manual paper-based record keeping, and fragmented communication channels. These decentralized models introduce operational inefficiencies, human errors, and significant security vulnerabilities. Customers routinely encounter lack of transparent pricing, unverified vehicle conditions, double-booking errors, and slow resolution of disputes or accidents. Conversely, independent vehicle owners lack a scalable, trusted platform to monetize their assets without incurring prohibitive marketing and operational overhead.

The proposed Online Vehicle Rental System (OVRs) establishes a centralized, high-throughput digital ecosystem that bridges the gap between individual vehicle owners, administrative managers, on-ground field personnel, and end-consumers. The system must guarantee transactional integrity under high concurrent loads, automate compliance validations (such as government identification and driving license authentications), enforce uniform business rules across payments, cancellations, and revenue splits, and maintain absolute physical transparency via real-time GPS tracking and internet-of-things (IoT) telemetry.

Detailed Problem Definition

Existing Problems in Legacy Systems

Manual vehicle rental practices rely on physical ledgers, spreadsheet updates, and verbal agreements, causing high operational latency, restricted business hours, and an inability to synchronize inventory across multiple hubs. Customers face hidden fees, arbitrary security deposits, and unverified wear-and-tear charges due to missing baseline condition records. Without distributed transaction isolation levels, concurrent manual bookings across fragmented channels trigger scheduling overlaps, double-booking errors, and severe revenue loss.

Offline cash and point-of-sale (POS) channels lack auditable records, making real-time payment reconciliation, instant cancellation refunds, and dynamic late-return charging impossible. The absence of real-time asset tracking prevents visibility into whether a vehicle is active, undergoing maintenance, or staged at a hub, leaving customers stranded upon arrival. Manual verification of driver licenses and identification documents for both self-drive clients and assigned drivers, is vulnerable to forgery and human error, exposing operators to legal non-compliance.

Furthermore, legacy setups lack automated driver allocation, intelligent recommendation algorithms, and structured promotional discount frameworks. When accidents occur, systems lack telemetry timelines, standardized damage reporting, and direct integration with insurance carriers. Independent vehicle vendors and owners encounter non-standardized onboarding sequences, untracked security deposits, and ambiguous revenue splits. Review systems are susceptible to manipulation through unverified edits and deletions, while single-node legacy architectures fail to scale under high concurrent loads during peak demand hours. Unclear boundaries also trigger frequent disputes between owners, vendors, and operators regarding maintenance liability and servicing responsibilities.

The Proposed Solution

The Online Vehicle Rental System (OVRs) addresses these issues by introducing a multi-tenant cloud application framework built on an ACID-compliant database architecture. It features integrated Automated Identity Verification APIs, real-time spatial query processing for tracking asset locations, an intelligent preference-based recommendation engine, and a deterministic state-machine workflow for handling bookings.

The system facilitates seamless interactions among Clients, Vendors, Drivers, and Administrators:

- **Clients:** Can securely register, search and apply advanced filters (price, vehicle type, brand, fuel type, seating capacity), receive personalized vehicle recommendations, choose between self-drive options (upon driving license

validation) or driver-allocated rentals, execute online payments with automated invoice generation, utilize promotional offers/coupons, track vehicles via GPS, and view complete booking histories.

- **Vendors & Owners:** Can register, upload vehicle documentation, pay security deposits, list and manage vehicle fleets, monitor rental histories, assign or request verified drivers, and receive automated revenue payouts.
- **Drivers:** Undergo automated credential verification, receive nearest-first booking assignments, and manage job acceptances/trips via specialized mobile interfaces.
- **On-Ground Personnel & Administrators:** All physical assets are staged at localized parking hubs where authorized workers run programmatic inspection checks to ensure no vehicle is re-rented without passing strict turnaround checks. Administrators maintain system-wide governance, oversee driver allotments, review flagged ratings, manage promotional campaigns, and monitor emergency SOS alerts.

By decoupling ownership, driver assignment, physical maintenance, on-ground staging, and digital orchestration, the platform creates an auditable ecosystem optimized for safety, scale, and high operational throughput.

SUMMARY OF REQUIREMENTS

1. ESSENTIAL SYSTEM FUNCTIONALITIES

- **User Onboarding & Verification:**
 - Provides multi-role registration workflows for Clients, Vendors, and Drivers.
 - Automates government ID and driving license authentications via third-party APIs.
 - Primary Actors: Client, Vendor, Driver.
- **Vehicle Inventory Management:**
 - Enables vendors to list, update, and manage multi-class vehicles (Cars, Two-wheelers, Bicycles).
 - Coordinates multi-hub staging and real-time status tracking.
 - Primary Actors: Vendor, Parking Worker.
- **Search & Smart Discovery:**
 - Allows clients to filter vehicles by price, fuel type, comfort rating, and proximity radius.
 - Integrates an AI-driven preference recommendation engine based on user history.

- Primary Actor: Client.
- **Transactional Booking Engine:**
 - Supports both immediate on-demand rentals and scheduled future reservations.
 - Uses atomic concurrency locking to prevent double-booking conflicts.
 - Primary Actors: Client, Core System.
- **Driver Allotment & Self-Drive Options:**
 - Dispatches verified, nearby drivers to bookings that request a driver.
 - Enables self-drive options for eligible clients following automated license checks.
 - Primary Actors: Client, Driver, Vendor.
- **Omnichannel Payment & Invoicing:**
 - Supports tokenized digital payments through UPI, Cards, Net Banking, and Wallets.
 - Generates automated PDF invoices upon transaction completion.
 - Executes an automated 80/20 revenue split between vendors and the platform.
 - Primary Actors: Client, Payment Gateway.
- **Telemetry & SOS Emergency System:**
 - Streams real-time GPS tracking coordinates for active rentals.
 - Features a one-touch emergency SOS dispatch button with evidence data locking.
 - Primary Actors: Client, Admin, Emergency Services.
- **Post-Rental Inspection Hold:**
 - Enforces a mandatory 30-minute maintenance and cleaning interlock upon vehicle return.
 - Re-inventories assets only after on-ground worker verification.
 - Primary Actors: Parking Worker, Core System.

2. FUNCTIONAL REQUIREMENTS (FR) SUMMARY

2.1 User Management & Eligibility

- **FR-1.1 (Multi-Role Onboarding):** The system must process role-specific registration workflows for Customers, Vendors, and Drivers.
- **FR-1.2 (Automated ID Validation):** The system must verify government IDs and date of birth using automated third-party APIs prior to account activation.
- **FR-1.3 (Motor Vehicle Eligibility):** The system must restrict motorized vehicle rentals to verified users aged 18+ holding a valid driving license.

- **FR-1.4 (Bicycle Eligibility):** The system must allow users aged 16+ to rent human-powered or electric bicycles without requiring a driver's license.

2.2 Inventory & Listing Governance

- **FR-2.1 (Asset Metadata Display):** Every vehicle listing must display photos, registration plate, hourly rate, fuel/battery capacity, mileage, and comfort rating.
- **FR-2.2 (Security Bond Processing):** Vendors must submit a refundable ₹5,000 security deposit per vehicle prior to listing activation.
- **FR-2.3 (Hub Staging Interlock):** On-ground workers must physically verify asset serial numbers and perform baseline inspections before staging vehicles at hubs.

2.3 Booking Engine & Concurrency Controls

- **FR-3.1 (Short-Duration VIN Locking):** Booking durations under 12 hours must lock a specific Vehicle Identification Number (VIN).
- **FR-3.2 (Atomic Collision Handling):** Overlapping simultaneous booking requests for a single asset must enforce row-level **SERIALIZABLE** database locking permitting only one transaction to clear while rejecting concurrent threads within 850 milliseconds.
- **FR-3.3 (Turnaround Inspection Lock):** Returned vehicles must automatically transition to an **IN_INSPECTION** state for a mandatory 30-minute cleaning and refueling window.

2.4 Payment, Fuel & Cancellation Guardrails

- **FR-4.1 (Payment Orchestration):** The platform must process payments via tokenized endpoints supporting UPI, Credit/Debit Cards, Net Banking, and Wallets.
- **FR-4.2 (Fault Recovery Cleanup):** Pre-payment drops or network timeouts must automatically release temporary reservation locks within 120 seconds.
- **FR-4.3 (15-Minute Grace Cancellation):** Full refunds must be issued for cancellations submitted within 15 minutes of initial booking payment.
- **FR-4.4 (Pro-Rata Fuel Billing):** Missing fuel or battery energy upon vehicle return must be automatically calculated and billed at current market rates plus a 15% service fee.
- **FR-4.5 (Automated Escrow Split):** Net rental fees must automatically split 80% to the Vendor/Owner escrow and 20% to the Platform treasury ledger.

2.5 Driver Management, Telemetry & Safety

- **FR-5.1 (Driver Allocation):** The system must auto-match verified available drivers to bookings requiring a driver based on location proximity.
- **FR-5.2 (SOS Disaster Dispatch):** Tapping the in-app SOS button must immediately broadcast live GPS coordinates, vehicle VIN, and user ID to admins and local emergency dispatch.
- **FR-5.3 (Accident Evidence Hold):** Logging an accident incident must lock all telemetry data from the preceding 30 minutes into a tamper-proof storage block for insurance claims.
- **FR-5.4 (Immutable Reviews):** User ratings and textual reviews must be append-only logs; manual deletion or modification by users or vendors is strictly prohibited.

3. NON-FUNCTIONAL REQUIREMENTS (NFR) SUMMARY

3.1 Performance & Scalability

- **NFR-1.1 (Throughput Capacity):** The system must sustain 5,000 read transactions per second and 1,500 write/booking updates per second with database lock latency under 250 milliseconds.
- **NFR-1.2 (Search Query Response):** Catalog search and filtering requests must render results within a maximum latency of 300 milliseconds under peak load.

3.2 Availability & Reliability

- **NFR-2.1 (System Uptime):** The core cloud platform must maintain 99.99% monthly operational uptime (excluding scheduled maintenance not exceeding 120 minutes per year).
- **NFR-2.2 (Circuit-Breaker Fault Tolerance):** Downstream payment or identity API outages must trigger graceful circuit-breaker degradation without crashing catalog browsing.

3.3 Security & Data Privacy

- **NFR-3.1 (Encryption Standards):** End-to-end data transmission must enforce TLS 1.3 encryption, and data at rest (ID hashes, payment tokens) must use AES-256 bit encryption.

- **NFR-3.2 (Role-Based Access Control):** All administrative and driver API operations must enforce Multi-Factor Authentication (MFA) and cryptographically signed JWT authorization tokens.

3.4 Auditability & Data Integrity

- **NFR-4.1 (Immutable Ledger):** All financial transactions, payouts, and balance updates must be recorded into a Write-Once-Read-Many (WORM) append-only database table.
- **NFR-4.2 (Telemetry Historical Retention):** Spatial GPS logs and device alert history must be preserved for a minimum of 365 days for legal and insurance compliance.